11 Chemistry
Unit 2 Practice Examination

Details	
Student Name:	Teacher Name:
Date Set: ____/____/26	Due Date: ____/____/26

Curriculum Assessed

On completion of Unit 2 the student should be able to measure and analyse chemicals in water: calculate solution concentrations, use volumetric analysis, apply the gas laws and gas stoichiometry, carry out calorimetry calculations for water, and explain and write equations for redox reactions, including metal displacement, corrosion and simple primary cells. This practice examination covers Unit 2.

Task Overview

Section A – Multiple Choice: 20 questions, 1 mark each (20 marks). Answer on the separate multiple-choice answer sheet.
Section B – Short Answer: 7 scenario-based questions (58 marks). Answer in this booklet.

Conditions

- Examination conditions (practice)
- VCE Chemistry Data Book provided
- Permitted materials: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator (no CAS).
- Not permitted: notes of any kind, blank paper, white-out, mobile phones, smart watches or any other electronic device.
- 10 mins reading time, 90 mins writing time

Marking

- 78 marks
- Marked according to teacher-made marking guide.
- Show all working in calculation questions; give final answers with units and an appropriate number of significant figures. Equations must be balanced and include states.

A+	A	B+	B	C+	C	D+	D	E+	E	NS
90-100%	80-89%	75-79%	70-74%	65-69%	60-64%	55-59%	50-54%	45-49%	40-44%	<40%

Total Marks:		Percentage:		Letter Grade:	

SECTION A – Multiple-choice questions (20 marks)

Answer **all** questions on the separate multiple-choice answer sheet by filling in the letter of the best response. Each question is worth 1 mark. No marks are deducted for incorrect answers. Do **not** write your Section A answers in this booklet.

Question 1

A student transferred 12.5 kJ of energy to 400.0 g of water, H_2O, at 19.0 °C. Assume that no energy was lost to the surroundings.

The final temperature of the water is closest to

A. 50.3 °C
B. 26.5 °C
C. 19.0 °C
D. 7.5 °C

Question 2

Which one of the following correctly gives the temperature and the pressure of standard laboratory conditions (SLC)?

	Temperature	Pressure
A.	0 °C	101.3 kPa
B.	25 °C	1.00 kPa
C.	298 °C	100 kPa
D.	25 °C	100 kPa

Question 3

Which one of the following human activities adds the greenhouse gas methane, CH_4, to the atmosphere?

A. raising large herds of cattle
B. burning petrol in car engines
C. making cement from limestone
D. burning coal in a power station

Question 4

Which one of the following solutions has the highest concentration in mol L^{-1}?

A. 5.00 g L^{-1} potassium bromide, KBr
B. 1.00% (m/v) glucose, $C_6H_{12}O_6$
C. 0.400% (m/v) sodium hydroxide, NaOH
D. 0.0600 M potassium chloride, KCl

Question 5

50.0 mL of 0.200 M sodium chloride solution, NaCl(aq), is mixed with 150.0 mL of 0.400 M sodium chloride solution.

The concentration of NaCl in the mixture is

A. 0.350 M
B. 0.300 M
C. 0.600 M
D. 0.0700 M

Question 6

A 5.00 L party balloon contains helium gas, He, at a pressure of 2.00 atm and a temperature of 25 °C.

The mass of helium in the balloon is closest to

- **A.** 0.0162 g
- **B.** 1.64 g
- **C.** 0.409 g
- **D.** 19.5 g

Question 7

At standard laboratory conditions (SLC), 1.00 L of an unknown gas has a mass of 1.13 g.

The gas is most likely to be

- **A.** neon, Ne
- **B.** carbon dioxide, CO_2
- **C.** methane, CH_4
- **D.** nitrogen, N_2

Question 8

Butane, C_4H_{10}, burns in a patio heater according to the equation

$2C_4H_{10}(g) + 13O_2(g) \rightarrow 8CO_2(g) + 10H_2O(g)$

The volume of oxygen, O_2, measured at SLC, needed to burn 29.0 g of butane completely is

- **A.** 12.4 L
- **B.** 161 L
- **C.** 1.91 L
- **D.** 80.6 L

Question 9

Calcium, Ca, reacts with sulfur, S, to form calcium sulfide, CaS:

$Ca(s) + S(s) \rightarrow CaS(s)$

A mixture of 4.01 g of calcium and 4.01 g of sulfur is heated until the reaction is complete.

Which one of the following is correct?

- **A.** Sulfur is the limiting reactant and 0.80 g of calcium remains.
- **B.** Both reactants are completely used up, because equal masses were used.
- **C.** Calcium is the limiting reactant and 0.80 g of sulfur remains.
- **D.** Calcium is the limiting reactant and 4.01 g of sulfur remains.

Question 10

A 25.00 mL aliquot of hydrochloric acid, HCl(aq), was titrated with 0.0450 M barium hydroxide solution, $Ba(OH)_2(aq)$. The average titre was 23.80 mL.

$Ba(OH)_2(aq) + 2HCl(aq) \rightarrow BaCl_2(aq) + 2H_2O(l)$

The concentration of the hydrochloric acid is

- **A.** 0.0857 M
- **B.** 0.0428 M
- **C.** 0.0214 M
- **D.** 0.0945 M

Question 11

Chlorine, Cl_2, dissolved in water is added to potassium bromide solution, KBr(aq). The ionic equation is

$Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$

Which one of the following statements about this reaction is correct?

A. Cl_2 is the reducing agent.

B. Br^- ions are reduced.

C. Cl_2 loses electrons.

D. Br^- ions are the reducing agent.

Question 12

In which one of the following species does manganese have an oxidation number of +6?

A. MnO_4^-

B. MnO_4^{2-}

C. MnO_2

D. Mn^{2+}

Question 13

In acidic conditions, nitrate ions, NO_3^-, can be reduced to nitrogen monoxide, NO.

In the balanced half-equation for this reduction, hydrogen ions, H^+, and electrons both appear on the reactant side. Their coefficients are

	Number of H^+	**Number of e^-**
A.	4	3
B.	2	3
C.	4	4
D.	4	5

Question 14

Which one of the following mixtures would result in a metal displacement reaction?

A. lead, Pb(s), in nickel(II) nitrate solution, $Ni(NO_3)_2(aq)$

B. silver, Ag(s), in copper(II) nitrate solution, $Cu(NO_3)_2(aq)$

C. nickel, Ni(s), in silver nitrate solution, $AgNO_3(aq)$

D. tin, Sn(s), in magnesium chloride solution, $MgCl_2(aq)$

Use the following information to answer Questions 15 and 16.

A student placed a clean iron, Fe, nail in each of four test tubes and left them for one week.

Tube	**Conditions**
1	dry air; anhydrous calcium chloride, $CaCl_2$, absorbs any water vapour; tube stoppered
2	water that had been boiled to remove dissolved oxygen, covered by a layer of oil
3	tap water, tube open to the air
4	nail painted all over, in tap water, tube open to the air

Question 15

In which tube would the most rust form?

A. tube 1

B. tube 2

C. tube 3

D. tube 4

Question 16

Aluminium, Al, reacts with chlorine gas, Cl_2, according to the equation

$2Al(s) + 3Cl_2(g) \rightarrow 2AlCl_3(s)$

Which one of the following is correct for this reaction?

	Oxidising agent	Total number of electrons transferred, as written
A.	Al	6
B.	Cl_2	6
C.	Cl_2	3
D.	$AlCl_3$	2

Question 17

The outlet of a bicycle pump is blocked, and the handle is pushed in until the volume of air inside the pump is halved. The temperature does not change.

The pressure of the air in the pump increases because the gas particles

A. collide with the walls of the pump more frequently.

B. move faster.

C. become smaller when they are squashed.

D. attract each other more strongly.

Question 18

Which one of the following samples of gas occupies the greatest volume at SLC?

A. 10.0 g of oxygen, O_2

B. 20.0 g of carbon dioxide, CO_2

C. 12.0 g of nitrogen, N_2

D. 1.00 g of hydrogen, H_2

Question 19

Simple galvanic (primary) cells can be made by connecting two metal half-cells, each with the metal in a 1.0 M solution of its own ions.

Using the electrochemical series in the Data Book, which pair of metals would produce the largest voltage?

A. zinc, Zn, and copper, Cu

B. iron, Fe, and lead, Pb

C. magnesium, Mg, and silver, Ag

D. nickel, Ni, and copper, Cu

Question 20

The same amount of energy is transferred to two beakers of water that start at the same temperature. Beaker X holds 100 g of water, H_2O, and beaker Y holds 400 g of water. The temperature of the water in beaker X rises by 12 °C.

The temperature rise in beaker Y is

A. 48 °C

B. 3.0 °C

C. 12 °C

D. 0.33 °C

END OF SECTION A

SECTION B – Short-answer questions (58 marks)

Answer **all** questions in the spaces provided. Show all working in calculation questions. Give final answers with units and appropriate significant figures. Equations must be balanced and include states, for example $H_2(g)$, NaCl(s).

Question 1 (10 marks)

Mia investigated the energy released when zinc powder, Zn, reacts with silver nitrate solution, $AgNO_3(aq)$.

She placed 60.0 mL of 0.400 M $AgNO_3(aq)$ in a polystyrene cup and recorded a temperature of 21.0 °C. She added 0.654 g of zinc powder and stirred. The highest temperature reached was 34.0 °C.

Assume that the solution has a density of 1.00 g mL^{-1} and the same specific heat capacity as water, H_2O.

The equation for the reaction is

$$Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$$

a) Write the reduction half-equation for this reaction, including states.

1 mark

b) Identify the reducing agent in this reaction.

1 mark

c) Show by calculation which reactant is limiting, and calculate the mass of silver nitrate, $AgNO_3$, left in excess when the reaction is complete.

3 marks

d) Calculate the energy released per mole of zinc that reacted, in kJ mol^{-1}.

3 marks

e) Mia's partner repeated the experiment with 120.0 mL of the same 0.400 M $AgNO_3(aq)$ and 0.654 g of zinc powder, starting at 21.0 °C. Explain whether the temperature rise would be larger than, smaller than or the same as Mia's.

2 marks

Question 2 (8 marks)

Huy grows lettuce hydroponically, with the roots sitting in a nutrient solution instead of soil. One of the nutrients is potassium nitrate, KNO_3.

His nutrient tank holds 20.0 L of solution, which must have a KNO_3 concentration of 0.0150 M.

a) Convert a concentration of 0.0150 M KNO_3 to g L^{-1}.

1 mark

b) Convert your answer to part a) to a concentration in % (m/v).

1 mark

c) Calculate the mass of KNO_3 that Huy would need to dissolve to make 20.0 L of 0.0150 M solution.

2 marks

d) Huy buys KNO_3 in 500 g bags. Calculate how many full 20.0 L tanks of 0.0150 M solution he can make from one bag.

2 marks

e) Calculate the mass of potassium ions, K^+, in a 250 mL sample of the 0.0150 M nutrient solution.

2 marks

Question 3 (8 marks)

The label on a window cleaner claims that it contains 4.5 g L^{-1} of ammonia, NH_3. Amira checked the claim by titrating 10.00 mL aliquots of the window cleaner with 0.1125 M hydrochloric acid, HCl(aq), from a burette. The mean of her concordant titres was 19.65 mL.

The equation for the reaction is

$$NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)$$

a) Before starting, Amira calculated the titre she would expect if the label were correct. Calculate this expected titre.

2 marks

b) Calculate the concentration of ammonia in the window cleaner, in mol L^{-1}.

2 marks

c) Determine whether Amira's result supports the claim on the label. Show your working.

2 marks

d) Ammonia is volatile: it slowly escapes from an open solution as a gas. Amira's partner suggests that this could explain the difference between the result and the label. Explain whether ammonia escaping from the bottle before the titration could account for the difference.

2 marks

Question 4 (10 marks)

On a dairy farm in Gippsland, cow manure is broken down in a sealed tank called a digester, producing biogas. The biogas is a mixture of methane, CH_4, and carbon dioxide, CO_2.

The biogas is stored in a 150 L steel tank at a pressure of 850 kPa and a temperature of 28.0 °C. Methane makes up 60.0% of the gas by amount (mol); the rest is carbon dioxide.

The biogas is burned in a generator to make electricity. The methane burns according to the equation below; the carbon dioxide already in the biogas passes through the generator unchanged.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

a) Methane and carbon dioxide are greenhouse gases. Name the other major greenhouse gas that is produced when methane burns in the generator, and give its formula.

1 mark

b) Calculate the total amount, in mol, of gas in the tank.

2 marks

c) Calculate the mass of methane in the tank.

2 marks

d) Calculate the total volume of carbon dioxide, measured at SLC, that leaves the generator when all the biogas in the tank is used. Include both the carbon dioxide formed from the methane and the carbon dioxide that was already in the biogas.

3 marks

e) Methane makes up 60.0% of the biogas by amount (mol). Calculate the percentage by mass of methane in the biogas.

2 marks

Question 5 (7 marks)

In a collision, a sensor in a car triggers the rapid decomposition of sodium azide, NaN_3 (M = 65.0 g mol^{-1}). The nitrogen gas, N_2, produced inflates the airbag.

$$2NaN_3(s) \rightarrow 2Na(s) + 3N_2(g)$$

a) A driver's airbag holds 60.0 L of nitrogen at a pressure of 1.30 atm and a temperature of 32.0 °C when it is fully inflated. Calculate the mass of sodium azide needed to inflate it.

4 marks

b) A student says that if the airbag were filled with 3.12 mol of argon, Ar, instead of nitrogen, it would need a larger bag because argon atoms are heavier. Explain why the student is wrong, in terms of the gas particles.

2 marks

c) Calculate the volume that the nitrogen in part a) would occupy at SLC.

1 mark

Question 6 (8 marks)

A water-treatment plant in a country town removes dissolved iron(II) ions, Fe^{2+}, from river water by oxidising them with potassium permanganate, $KMnO_4$ (M = 158.0 g mol^{-1}).

In a laboratory model of the process, carried out in acidic conditions, permanganate ions, MnO_4^-, are reduced to manganese(II) ions, Mn^{2+}, and the iron(II) ions are oxidised to iron(III) ions, Fe^{3+}.

a) Write the oxidation half-equation, including states.

1 mark

b) Write the balanced reduction half-equation for permanganate ions in acidic conditions, including states.

2 marks

c) Write the balanced overall equation for the reaction, including states.

1 mark

d) Identify the reducing agent in this reaction.

1 mark

e) A 5000 L storage tank of river water contains 3.50 mg L^{-1} of Fe^{2+} ions. Calculate the mass of $KMnO_4$ needed to oxidise all of the Fe^{2+} in the tank.

3 marks

Question 7 (7 marks)

Zoe built four simple galvanic cells. In each cell, a strip of a metal X in a 1.0 M solution of its nitrate was connected by a salt bridge to a strip of copper, Cu, in 1.0 M copper(II) nitrate solution, $Cu(NO_3)_2(aq)$. The nitrates used were aluminium nitrate, $Al(NO_3)_3$, iron(II) nitrate, $Fe(NO_3)_2$, tin(II) nitrate, $Sn(NO_3)_2$, and silver nitrate, $AgNO_3$.

Zoe measured the voltage of each cell at 25 °C and used the voltmeter to find which electrode was positive. Her results are shown below.

Metal X	**aluminium, Al**	**iron, Fe**	**tin, Sn**	**silver, Ag**
Voltage of cell (V)	1.95	0.75	0.46	0.44
Polarity of the copper electrode	positive	positive	positive	negative

a) State one function of the salt bridge in Zoe's cells.

1 mark

b) Write the balanced overall equation, including states, for the reaction in the silver–copper cell.

2 marks

c) Write the oxidation half-equation for the reaction at the anode of the aluminium–copper cell, including states.

1 mark

d) Zoe did not build an aluminium–iron cell. Use her results to predict the voltage of an aluminium–iron cell, and explain your prediction.

2 marks

e) Use Zoe's results to list the five metals (aluminium, iron, tin, copper and silver) in order of their strength as reducing agents, strongest first.

1 mark

END OF SECTION B

Additional space

Clearly number any response written here.

END OF EXAMINATION